Problem · Unit 4 · 10 points · 60 min · Standard exam level

***P06 — Precessing orbit under an attractive inverse-cube perturbation: Binet equation, circular orbits and apsidal advance***

*A particle $P$ of mass $m$ moves about a fixed centre $O$ (inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$) under the single force*

$$\mathbf{F} = -\left(\frac{m\mu}{r^2} + \frac{m\varepsilon}{r^3}\right)\mathbf{e}_r$$

*where $r = |\mathbf{OP}|$, $\mathbf{e}_r = \mathbf{OP}/r$, $\mu > 0$ is the gravitational parameter of the centre and $\varepsilon > 0$ (units $m^4\,s^{-2}$) measures a small attractive perturbation (a model perturbation added to the Newtonian attraction). Once the motion is shown to be planar, use polar coordinates $(r, \theta)$ in the plane of the orbit, with $\theta$ measured counter-clockwise from a fixed reference line through $O$, the unit vectors $\mathbf{e}_r$, $\mathbf{e}_\theta$ ($\mathbf{e}_\theta$ points towards increasing $\theta$) and $h = r^2\dot{\theta}$.*

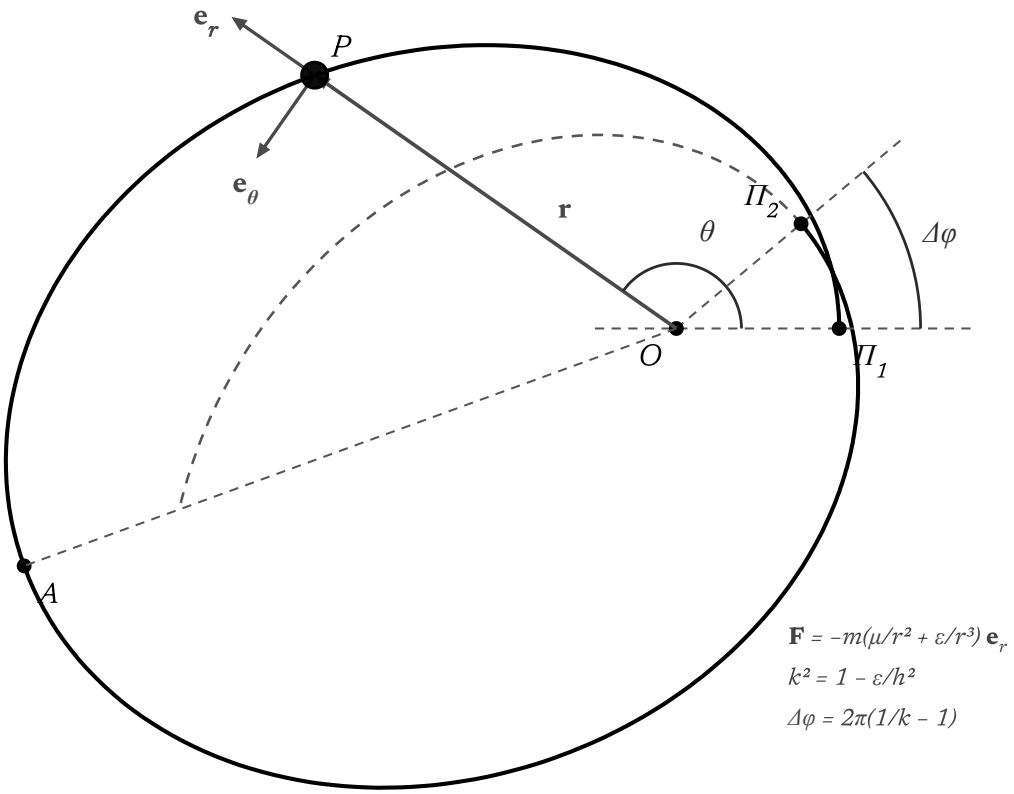



*Figure 1. Orbit of $P$ about $O$ for $\varepsilon=0.19\,h^2$, $e = 0.62$: one radial cycle from the perigee $\Pi_1$ through the apogee $A$ to the next perigee $\Pi_2$ (solid) and the start of the next cycle (dashed); $r$, $\theta a$, the polar unit vectors at $P$ and the apsidal advance $\Delta\phi$.*

*A perigee is a point of minimum $r$ and an apogee a point of maximum $r$. In the numerical part the reference line $\theta = 0$ passes through a perigee.*

## Setup

- Data: $m$, $\mu$, $\varepsilon$ and the initial conditions. Numerical case (part (f)): $\mu = 4.0 \times 10^{14}$ m$^3$ s$^{-2}$; at $t = 0$ the particle is at the perigee $r_p = 8.0 \times 10^6$ m with the speed $v_p = 1.0 \times 10^4$ m/s perpendicular to $\mathbf{OP}$; $\varepsilon = 0.19\, h^2$.
- Hypotheses: point particle, $O$ fixed and $S_0$ inertial, no other force.
- Frames: $S_0$ (inertial, Cartesian) and the polar basis $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ attached to $P$, with $\mathbf{e}_r = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$ and $\mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$ ($\mathbf{i}, \mathbf{j}$ in the plane of the orbit, $\mathbf{k}$ along $\mathbf{H}_O$, see (a)).
- Change of basis:

$$\left(\mathbf{e}_r\ \mathbf{e}_\theta\ \mathbf{k}\right) = [R]\left(\mathbf{i}\ \mathbf{j}\ \mathbf{k}\right), \qquad [R] = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

with rows of unit length ($\cos^2\theta + \sin^2\theta = 1$) and mutually orthogonal ($-\cos\theta\sin\theta + \sin\theta\cos\theta = 0$), so $[R][R]^T = I$, and $\det[R] = \cos^2\theta + \sin^2\theta = +1$: the polar basis is right-handed, $\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{k}$.

## Part (a) · 1.5 points

*Planar motion and conservation of angular momentum*

*Show that the angular momentum $\mathbf{H}_O$ of $P$ about $O$ is constant, that the motion takes place in a fixed plane through $O$, and that $h = r^2\dot\theta$ is a constant of the motion. How many degrees of freedom remain?*

**Step 1. Torque of a central force**

$\mathbf{F}$ is parallel to $\mathbf{r} = r\,\mathbf{e}_r$, so

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = -\left(\frac{m\mu}{r^2} + \frac{m\varepsilon}{r^3}\right) r\,(\mathbf{e}_r \times \mathbf{e}_r) = \mathbf{0}$$

By the angular momentum theorem about a fixed point (Notes §4.8), $\dot{\mathbf{H}}_O = \mathbf{M}_O = \mathbf{0}$, so $\mathbf{H}_O = m\,\mathbf{r} \times \mathbf{v}$ is a constant vector.

**Step 2. Plane of the motion**

For every $t$, $\mathbf{r}\cdot\mathbf{H}_O = m\,\mathbf{r}\cdot(\mathbf{r}\times\mathbf{v}) = 0$ (a triple product with a repeated vector). The position is always orthogonal to the fixed vector $\mathbf{H}_O$: $P$ moves in the fixed plane through $O$ orthogonal to $\mathbf{H}_O$ (if $\mathbf{H}_O = \mathbf{0}$ the motion is radial, along a fixed line). Choose $\mathbf{k}$ along $\mathbf{H}_O$ (Notes §4.8.1).

**Step 3. The constant $h$**

In the plane, $\mathbf{r} = r\,\mathbf{e}_r$ and, with $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ (chain rule on $\mathbf{e}_r(\theta(t))$),

$$\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{H}_O = m\,r\,\mathbf{e}_r \times \left(\dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta\right) = mr^2\dot\theta\,\mathbf{k}$$

So $h = |\mathbf{H}_O|/m = r^2\dot\theta$ is constant (twice the areal velocity). The particle has 3 degrees of freedom in space; the plane is fixed by the initial conditions, and in it 2 remain, $r$ and $\theta$, linked by the first integral $h$.

KEY RESULTS

- $\mathbf{M}_O = \mathbf{r}\times\mathbf{F} = \mathbf{0}$, so $\mathbf{H}_O = m\,\mathbf{r}\times\mathbf{v}$ is constant.
- $\mathbf{r}\cdot\mathbf{H}_O = 0$: $P$ stays in the plane through $O$ orthogonal to $\mathbf{H}_O$; in it $\mathbf{H}_O = mr^2\dot\theta\,\mathbf{k}$, so $h = r^2\dot\theta =$ const.
- 2 degrees of freedom, $(r,\theta)$, with one first integral ($h$).

## Part (b) · 1.5 points

*Equations of motion and energy*

*Write the equations of motion in polar components. Show that $\mathbf{F}$ is conservative, find its potential energy, and write the energy integral as a one-dimensional problem in $r$ with an effective potential $U_{eff}(r)$.*

**Step 1. Acceleration in polar components**

Differentiating $\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta$ with the product rule and $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$, $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$:

$$\mathbf{a} = \ddot r\,\mathbf{e}_r + \dot r\dot\theta\,\mathbf{e}_\theta + (\dot r\dot\theta + r\ddot\theta)\,\mathbf{e}_\theta - r\dot\theta^2\,\mathbf{e}_r = (\ddot r - r\dot\theta^2)\,\mathbf{e}_r + (2\dot r\dot\theta + r\ddot\theta)\,\mathbf{e}_\theta$$

and $2\dot r\dot\theta + r\ddot\theta = \frac{1}{r}\frac{d}{dt}(r^2\dot\theta)$ (product rule: $\frac{d}{dt}(r^2\dot\theta) = 2r\dot r\dot\theta + r^2\ddot\theta$).

**Step 2. Newton's second law**

$m\mathbf{a} = \mathbf{F}$ projected on $\mathbf{e}_r$ and $\mathbf{e}_\theta$ (Notes §8.3):

$$\ddot r - r\dot\theta^2 = -\frac{\mu}{r^2} - \frac{\varepsilon}{r^3}, \qquad \frac{1}{r}\frac{d}{dt}(r^2\dot\theta) = 0$$

The second equation is the conservation of $h$ of part (a).

**Step 3. Potential energy**

$\mathbf{F} = f(r)\,\mathbf{e}_r$ with $f(r) = -m\mu/r^2 - m\varepsilon/r^3$ depends only on $r$, so $\mathbf{F} = -\nabla V$ with $V'(r) = -f(r)$ (Notes §4.4). Choosing $V \to 0$ as $r \to \infty$, $V(r) = -\displaystyle\int_\infty^r f(r')\,dr'$. An antiderivative of $-f(r') = \dfrac{m\mu}{r'^2} + \dfrac{m\varepsilon}{r'^3}$ is $G(r') = -\dfrac{m\mu}{r'} - \dfrac{m\varepsilon}{2r'^2}$, and Barrow's rule with $G(r') \to 0$ as $r' \to \infty$ gives

$$V(r) = \int_\infty^r \Big(\frac{m\mu}{r'^2} + \frac{m\varepsilon}{r'^3}\Big)dr' = \Big[-\frac{m\mu}{r'} - \frac{m\varepsilon}{2r'^2}\Big]_\infty^r = \Big(-\frac{m\mu}{r} - \frac{m\varepsilon}{2r^2}\Big) - 0 = -\frac{m\mu}{r} - \frac{m\varepsilon}{2r^2}$$

Check: $-V'(r) = -\frac{m\mu}{r^2} - \frac{m\varepsilon}{r^3} = f(r)$.

**Step 4. Energy integral and effective potential**

The only force is conservative, so $E = \frac{1}{2}m|\mathbf{v}|^2 + V$ is constant (Notes §4.5), with $|\mathbf{v}|^2 = \dot r^2 + r^2\dot\theta^2 = \dot r^2 + h^2/r^2$:

$$\frac{E}{m} = \tfrac{1}{2}\dot r^2 + \frac{h^2}{2r^2} - \frac{\mu}{r} - \frac{\varepsilon}{2r^2} = \tfrac{1}{2}\dot r^2 + U_{eff}(r), \qquad U_{eff}(r) = \frac{h^2 - \varepsilon}{2r^2} - \frac{\mu}{r}$$

The perturbation enters exactly as a reduction of the centrifugal barrier, $h^2 \to h^2 - \varepsilon$.

KEY RESULTS

- $\ddot r - r\dot\theta^2 = -\dfrac{\mu}{r^2} - \dfrac{\varepsilon}{r^3}$ and $\dfrac{d}{dt}(r^2\dot\theta) = 0$.
- $V = -\dfrac{m\mu}{r} - \dfrac{m\varepsilon}{2r^2}$.
- $\dfrac{E}{m} = \tfrac{1}{2}\dot r^2 + U_{eff}(r)$, $U_{eff} = \dfrac{h^2 - \varepsilon}{2r^2} - \dfrac{\mu}{r}$.

## Part (c) · 1.5 points

*Circular orbit and its stability*

*For a given $h$, find the radius $r_c$ and the speed of the circular orbit. Under what condition does it exist? Is it stable against small radial perturbations? Compare the frequency of the small radial oscillations with the angular rate on the circle.*

**Step 1. Equilibrium of the radial problem**

$$U'_{eff}(r) = -\frac{h^2 - \varepsilon}{r^3} + \frac{\mu}{r^2} = 0 \implies r_c = \frac{h^2 - \varepsilon}{\mu}$$

A positive radius requires $h^2 > \varepsilon$. If $h^2 \le \varepsilon$ the barrier term is non-positive, $U_{eff}$ decreases monotonically towards $r \to 0$, and the particle with $\dot r < 0$ falls into the centre: no circular orbit exists.

**Step 2. Speed**

On the circle $\dot r = 0$, so $v_c = r_c\dot\theta = h/r_c$. Equivalently, the radial equation with $\ddot r = 0$ gives $r_c\dot\theta^2 = \mu/r_c^2 + \varepsilon/r_c^3$, i.e. $v_c^2 = \mu/r_c + \varepsilon/r_c^2$; with $h^2 = \mu r_c + \varepsilon$ both forms agree.

**Step 3. Stability**

$$U''_{eff}(r) = \frac{3(h^2 - \varepsilon)}{r^4} - \frac{2\mu}{r^3} \implies U''_{eff}(r_c) = \frac{3\mu r_c}{r_c^4} - \frac{2\mu}{r_c^3} = \frac{\mu}{r_c^3} > 0$$

$r_c$ is a minimum of $U_{eff}$: the circular orbit is stable. Linearising $\ddot r = -U'_{eff}(r)$ about $r_c$ with $\rho = r - r_c$ gives $\ddot\rho + U''_{eff}(r_c)\rho = 0$, so the radial oscillations have $\omega_r^2 = \mu/r_c^3$.

**Step 4. Radial frequency versus angular rate**

On the circle $\dot\theta_c = h/r_c^2$, so $\dot\theta_c^2 = h^2/r_c^4$ and

$$\frac{\omega_r^2}{\dot\theta_c^2} = \frac{\mu r_c}{h^2} = \frac{h^2 - \varepsilon}{h^2} = 1 - \frac{\varepsilon}{h^2}$$

For $\varepsilon = 0$ (Kepler) the ratio is $1$: $r$ returns to its minimum exactly once per turn and the slightly perturbed orbit closes (an ellipse). For $\varepsilon > 0$ the radial oscillation is slower than the rotation, so the minimum of $r$ is reached **after** more than one turn: the perigee advances. Part (e) makes this exact for any amplitude.

KEY RESULTS

- $r_c = \dfrac{h^2 - \varepsilon}{\mu}$, it exists only if $h^2 > \varepsilon$; $v_c^2 = \dfrac{h^2}{r_c^2} = \dfrac{\mu}{r_c} + \dfrac{\varepsilon}{r_c^2}$.
- $U''_{eff}(r_c) = \mu / r_c^3 > 0$: stable. $\omega_r^2 = \mu / r_c^3,\ \omega_r / \dot{\theta}_c = \sqrt{1 - \varepsilon / h^2} < 1$.

## Part (d) · 2 points

*Binet equation and the orbit*

*With $u = 1/r$ and $\theta$ as the independent variable, derive the Binet equation of this force field, solve it with the perigee at $\theta = 0$, and give the perigee and apogee radii.*

**Step 1. First derivative**

$r = 1/u$ with $u = u(\theta(t))$. Chain rule:

$$\dot{r} = \frac{d}{dt}\left(\frac{1}{u}\right) = -\frac{1}{u^2}\frac{du}{d\theta}\dot{\theta} = -\frac{1}{u^2}u'\, hu^2 = -h\, u'$$

using $\dot{\theta} = h/r^2 = hu^2$.

**Step 2. Second derivative**

$h$ is constant, so again by the chain rule

$$\ddot{r} = \frac{d}{dt}(-h\, u') = -h\frac{du'}{d\theta}\dot{\theta} = -h\, u''\, hu^2 = -h^2 u^2 u''$$

**Step 3. Radial equation in $u$**

Substitute $\ddot{r}$, $r\dot{\theta}^2 = \frac{1}{u}\, h^2 u^4 = h^2 u^3$, $\mu/r^2 = \mu u^2$ and $\varepsilon/r^3 = \varepsilon u^3$ in the radial equation of (b):

$$-h^2 u^2 u'' - h^2 u^3 = -\mu u^2 - \varepsilon u^3$$

Dividing by $-h^2 u^2$ (Notes §8.3, Eq. 8.12 with the extra term):

$$u'' + \left(1 - \frac{\varepsilon}{h^2}\right)u = \frac{\mu}{h^2}$$

**Step 4. Solution**

With the scalar $k^2 = 1 - \varepsilon/h^2 > 0$ (not the unit vector $\mathbf{k}$; the condition $h^2 > \varepsilon$ of (c)), this is a forced harmonic oscillator in $\theta$ with "frequency" $k$. A particular solution is the constant $u_0 = \mu/(h^2k^2)$; the homogeneous solution is $C\cos(k\theta - \theta_0)$. Choosing $\theta = 0$ at a perigee (maximum of $u$) gives $\theta_0 = 0$ and $C \geq 0$; write $C = u_0 e$:

$$u = \frac{\mu}{h^2k^2}\big(1 + e\cos k\theta\big), \qquad r = \frac{r_c}{1 + e\cos k\theta}, \qquad r_c = \frac{h^2k^2}{\mu} = \frac{h^2 - \varepsilon}{\mu}$$

Check: $u'' = -k^2u_0 e\cos k\theta$ and $u'' + k^2u = k^2u_0 = \mu/h^2$.

**Step 5. Apsides**

$r$ is minimum where $\cos k\theta = 1$ ($\theta = 0$) and maximum where $\cos k\theta = -1$ ($k\theta = \pi$):

$$r_p = \frac{r_c}{1+e}, \qquad r_a = \frac{r_c}{1-e} \quad (0 \leq e < 1)$$

The shape looks like a conic with semi-parameter $r_c$, but the angle that appears is $k\theta$, not $\theta$: the apogee is reached at $\theta = \pi/k > \pi$.

KEY RESULTS

- $u'' + \left(1 - \dfrac{\varepsilon}{h^2}\right)u = \dfrac{\mu}{h^2}$.
- $u = \dfrac{\mu}{h^2k^2}\big(1 + e\cos k\theta\big)$, i.e. $r = \dfrac{r_c}{1 + e\cos k\theta}$ with $r_c = \dfrac{h^2 - \varepsilon}{\mu} = \dfrac{h^2k^2}{\mu}$.
- $r_p = \dfrac{r_c}{1+e}$ at $\theta = 0$; $r_a = \dfrac{r_c}{1-e}$ at $k\theta = \pi$ (bounded orbit if $0 \leq e < 1$).

## Part (e) · 1.5 points

*Apsidal precession*

*Find the angle swept between two consecutive perigees and the advance $\Delta\phi$ of the perigee per radial cycle; give its first-order form for $\varepsilon \ll h^2$. Is the orbit closed? Is the time between two perigees the time the particle needs to turn $360^\circ$?*

**Step 1. Angle between perigees**

Perigees are at $\cos k\theta = 1$, i.e. $k\theta = 0, 2\pi, 4\pi, \dots$ Two consecutive ones are separated by $\theta = 2\pi/k$, and since $k < 1$ this is more than one turn. The advance per radial cycle is

$$\Delta\phi = \frac{2\pi}{k} - 2\pi = 2\pi\Big(\frac{1}{\sqrt{1-\varepsilon/h^2}} - 1\Big)$$

positive: the perigee advances in the sense of the motion.

**Step 2. Small perturbation**

With $x = \varepsilon/h^2 \ll 1$ and the binomial series $(1-x)^{-1/2} = 1 + \frac{1}{2}x + O(x^2)$:

$$\Delta\phi = 2\pi\Big(\tfrac{1}{2}x + O(x^2)\Big) \approx \frac{\pi\varepsilon}{h^2}$$

**Step 3. Closed or not**

The orbit closes if after $q$ radial cycles the particle has made an integer number $p$ of turns: $q\,(2\pi/k) = 2\pi p$, i.e. $k = q/p$ rational. Otherwise the perigee directions never repeat and the trajectory densely fills the annulus between $r_p$ and $r_a$.

**Step 4. Time between perigees (the trap)**

Substituting $\dot\theta = h/r^2$ in the radial equation of (b):

$$\ddot r = \frac{h^2}{r^3} - \frac{\mu}{r^2} - \frac{\varepsilon}{r^3} = \frac{\tilde h^2}{r^3} - \frac{\mu}{r^2}, \qquad \tilde h^2 = h^2 - \varepsilon$$

This is the radial equation of a Kepler orbit with angular momentum $\tilde h$ and the same $r_p, r_a$, so $r(t)$ is periodic with the Kepler period of semi-major axis $a = (r_p + r_a)/2$ (Notes §8.4, Eq. 8.18):

$$T_r = 2\pi\sqrt{\frac{a^3}{\mu}}$$

During $T_r$ the angle advances by $2\pi/k$, not $2\pi$: $\dot\theta = h/r^2 = (h/\tilde h)\,(\tilde h/r^2)$ and $h/\tilde h = 1/k$. The time to turn $360^\circ$ and the time between perigees are different quantities; equating them is the classic error. The energy also matches: at $r_p$, $\frac{E}{m} = U_{eff}(r_p) = -\frac{\mu}{2a}$.

KEY RESULTS

- Angle between perigees $2\pi/k$; $\Delta\phi = 2\pi\Big(\dfrac{1}{\sqrt{1-\varepsilon/h^2}} - 1\Big) \approx \dfrac{\pi\varepsilon}{h^2}$.
- Closed only if $k$ is rational; for irrational $k$ the orbit fills the annulus $r_p \le r \le r_a$ (a rosette).
- Time between perigees: the radial period $T_r = 2\pi\sqrt{a^3/\mu}$, $2a = r_p + r_a$; in that time $\theta$ advances $2\pi/k$, **not** $2\pi$.

## Part (f) · 2 points

*Numerical case*

*With the data of the Setup ($\mu = 4.0 \times 10^{14}\ m^3\ s^{-2}$, $r_p = 8.0 \times 10^6\ m$, $v_p = 1.0 \times 10^4\ m/s$ perpendicular to* **OP**, *$\varepsilon = 0.19\,h^2$), compute $h$, $\varepsilon$, $k$, $r_c$, $e$, $r_a$, $E/m$, the apsidal advance (exact and first-order), and the time between two perigees. Comment on the value of $v_p$.*

**Step 1. Constants of the motion**

$h = r_p v_p = 8.0 \times 10^6 \times 1.0 \times 10^4 = 8.0 \times 10^{10}\ \mathrm{m^2/s}$, $h^2 = 6.4 \times 10^{21}\ \mathrm{m^4\ s^{-2}}$, $\varepsilon = 0.19\,h^2 = 1.216 \times 10^{21}\ \mathrm{m^4\ s^{-2}}$ and $k = \sqrt{1 - 0.19} = \sqrt{0.81} = 0.9$.

**Step 2. Orbit**

$$r_c = \frac{h^2 - \varepsilon}{\mu} = \frac{0.81 \times 6.4 \times 10^{21}}{4.0 \times 10^{14}} = 1.296 \times 10^7\ \mathrm{m}$$

At $\theta = 0$, $r = r_p$: $r_p = r_c/(1+e) \Rightarrow e = r_c/r_p - 1 = 1.62 - 1 = 0.62$. Then $r_a = r_c/(1-e) = 1.296 \times 10^7/0.38 = 3.411 \times 10^7$ m and $a = (r_p + r_a)/2 = 2.105 \times 10^7$ m.

**Step 3. Energy**

$$\frac{E}{m} = \frac{v_p^2}{2} - \frac{\mu}{r_p} - \frac{\varepsilon}{2r_p^2} = 5.0 \times 10^7 - 5.0 \times 10^7 - \frac{1.216 \times 10^{21}}{1.28 \times 10^{14}} = -9.50 \times 10^6\ \mathrm{J/kg}$$

in agreement with $-\mu/(2a) = -4.0 \times 10^{14}/(4.211 \times 10^7) = -9.50 \times 10^6$ J/kg. Note that $\tfrac{1}{2}v_p^2 = \mu/r_p$: $v_p$ is exactly the **escape speed** of the unperturbed problem. Without the perturbation the orbit would be a parabola; the extra attraction makes it bound.

**Step 4. Precession and times**

$$\Delta\phi = 2\pi\left(\frac{1}{0.9} - 1\right) = \frac{2\pi}{9} = 40^\circ, \qquad \frac{\pi\varepsilon}{h^2} = 0.19\,\pi = 0.597\ \text{rad} = 34.2^\circ$$

The first-order formula underestimates by 15 %: $\varepsilon/h^2 = 0.19$ is not small. The apogee is at $\theta = \pi/k = 200^\circ$ and the next perigee at $\theta = 400^\circ$. Since $k = 9/10$, the orbit closes after 9 radial cycles and 10 turns.

$$T_r = 2\pi\sqrt{\frac{(2.105\times10^7)^3}{4.0\times10^{14}}} = 3.035\times10^4\ \text{s} \approx 8.43\ \text{h}$$

A numerical integration of the equations of (b) from the perigee confirms that $r$ is minimum again after $3.035\times10^4$ s with $\theta = 400.0^\circ$.

KEY RESULTS

- $h = 8.0\times10^{10}$ m$^2$/s, $\varepsilon = 1.216\times10^{21}$ m$^4$ s$^{-2}$, $k = 0.9$.
- $r_c = 1.296\times10^7$ m, $e = 0.62$, $r_a = 3.411\times10^7$ m, $a = 2.105\times10^7$ m, $E/m = -9.50\times10^6$ J/kg.
- $\Delta\phi = 40^\circ$ exactly ($2\pi/9$); first-order estimate $34.2^\circ$; apogee at $\theta = 200^\circ$; $T_r = 3.035\times10^4$ s.

## Interpretation and checks

- **Dimensional check:** $\mu/r^2$ and $\varepsilon/r^3$ are accelerations (m$^3$ s$^{-2}$/m$^2$ and m$^4$ s$^{-2}$/m$^3$); $\varepsilon/h^2$ is dimensionless (m$^4$ s$^{-2}$ over m$^4$ s$^{-2}$), so $k$ and $\Delta\phi$ are pure numbers; $r_c = (h^2 - \varepsilon)/\mu$ has m$^4$ s$^{-2}$/(m$^3$ s$^{-2}$) = m; $\sqrt{a^3/\mu}$ has s.
- **Limits:** $\varepsilon \to 0$ gives $k = 1$, $\Delta\phi = 0$, the conic $r = (h^2/\mu)/(1 + e\cos\theta)$ of Notes Eq. 8.14 and $T_r$ equal to the orbital period. $\varepsilon \to h^2$ gives $k \to 0$: the barrier disappears, $\Delta\phi \to \infty$ and $r_c \to 0$ (the particle spirals in).
- **Common mistakes in the exam:** (1) dropping the $\varepsilon u^3$ term when changing to $u$, or writing it on the wrong side, which changes the sign of the precession; (2) treating the orbit as a closed ellipse with $\theta$ in place of $k\theta$ (apogee at 180°); (3) taking the time between perigees as the time for one turn, or dividing $\Delta\phi$ by the period of a Kepler orbit with $h$ instead of $\tilde{h}$; (4) using the first-order $\pi\varepsilon/h^2$ when $\varepsilon/h^2$ is not small.